

Student Guide — middle / module-01- what-is-iks

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — Module 1 (Middle)

Name: _____ Class: _____

Day 1 — What is IKS?

My working definition (in my own words):

Day 2 — The Four Vedas

Veda	What it contains
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R̥g

Yajur

Sāma

Atharva

Four layers within each Veda: 1. Saṃhitā — _____ 2. Brāhmaṇa — _____ 3. Āraṇyaka — _____ 4. Upaniṣad — _____

Day 3 — The Six Vedangas

1. Śikṣā – phonetics
2. Kalpa – ritual
3. Vyākaraṇa – grammar (Pāṇini)
4. Nirukta – etymology
5. Chandas – prosody
6. Jyotiṣa – astronomy

Day 4 — The Four Upavedas

#	Name	Domain
1	Āyurveda	medicine

#	Name	Domain
2	Dhanurveda	martial
3	Gandharvaveda	music
4	Arthaśāstra	governance

Day 5 — Quiz score: _____ / 12

Day 6 — Indian Math

Three contributions I now know: 1. _____ (date: _____) 2. _____ (date: _____) 3. _____ (date: _____)

Mādhava series for π — my 10-term computation: $\pi \approx$ _____

Day 7 — Indian Astronomy

Aryabhata proposed: _____ Aryabhata did NOT propose: _____

Days 8–10 — Project

Topic: _____ Named practitioner: _____ Specific claim: _____ Source: _____ Limitation: _____

Day 10 — Reflection

1. What did I set out to research? _____
2. What surprised me? _____
3. With 3 more days, deeper research on: _____

Day 1 — The Hook: What Does “Indian Knowledge” Mean?

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

By the end of this lesson, students will give an open-ended definition of “Indian Knowledge Systems” that mentions at least three distinct domains (e.g. mathematics, medicine, philosophy).

Materials

- Whiteboard
- Sticky notes (3 per student)
- Wall poster: “Indian contributions you’ve encountered THIS WEEK” (to be filled together)

Vocabulary

- *vidyā* — “knowledge, learning”
- *śāstra* — “treatise; systematic body of knowledge”
- *jñāna* — “knowledge as understanding” (different shade than *vidyā*)

Flow

Warm-up (5 min)

Question on board: “*What’s one Indian-origin idea you’ve encountered this week, outside this class?*”

Students write on sticky notes. Post on the wall. Read aloud the variety: yoga, zero (in math), turmeric (kitchen), an Indian-named star (Rigel? actually Arabic — but the question is fertile), namaste at the start of a class, an Ayurvedic ingredient on a shampoo bottle.

Core (20 min) — A working definition

1. **The question:** “*If someone said to you ‘tell me about Indian knowledge,’ what would you say?*”

- Take 3–4 student answers. Don’t correct. Note variety.

2. **Build a working definition** on the board:

Indian Knowledge Systems (IKS) = the intellectual traditions developed on the Indian subcontinent over ~3000 years — including science, mathematics, astronomy, linguistics, philosophy, medicine, music, architecture, and more.

3. **Three things IKS is NOT:**

- **NOT only religion.** (Some IKS is religious; much is not.)
- **NOT magical thinking dressed up as science.** (Real claims, real methods.)
- **NOT a competitor to modern science.** (Both can be true at the same time.)

4. **Three things IKS IS:**

- A real body of intellectual work with specific authors, dates, and texts.
- Sometimes overlapping with, sometimes distinct from, religious tradition.
- Worth studying as history of human inquiry.

5. **One specific example to anchor today:**

- The Indian invention of **zero** as a number (not just placeholder) is documented in Bakhshali manuscript (~7th c. CE), and used in Bhāskara II’s *Līlāvati* (12th c. CE). It travelled via Arab mathematicians to Europe by ~12th c.
- That’s a real, dated, sourceable contribution.

Activity (15 min) — Inventory

Each student writes on 3 sticky notes: 1. One thing they **ALREADY KNEW** about IKS (before today). 2. One thing they **WANT TO LEARN** about IKS in this module. 3. One thing they're **SKEPTICAL** of (any claim about IKS they doubt).

Post on the wall. Cluster by theme during break.

Wrap-up (5 min)

- Note out loud: “By Day 10 you’ll each give a 5-minute presentation on **ONE** specific Indian contribution. Start thinking about which strand interests you most.”
- Preview Day 2: “*Tomorrow — the Vedas. Not the religious part you might have heard about. The intellectual part.*”

Student-facing content

What is IKS?

Indian Knowledge Systems = the intellectual traditions of the Indian subcontinent (~3000 years).

Includes: mathematics, astronomy, medicine, linguistics, philosophy, music, architecture.

One example: **the number zero**. First documented as a number (not just a placeholder) in the **Bakhshali manuscript** (~7th century CE). Used in Bhāskara II’s *Līlāvātī* (1150 CE). Reached Europe by the 12th century via Arab mathematicians.

Homework

- Read the project brief ([assessments/project-brief.md](#)). Think about which strand of IKS you might pick.

Differentiation

- **Tier up:** Find **ONE** thing you previously believed about IKS that turned out to be wrong (or oversold). Bring tomorrow.
- **Tier down:** Just remember the working definition: *intellectual traditions of the Indian subcontinent*.

Teacher notes

- The “what would you say?” prompt often reveals real diversity — some students will say “religion,” some “yoga,” some “math.” All are partly right.
- The “**NOT** only religion” framing is important early. Many students conflate IKS with Hinduism. They overlap; they are not identical.

Citation(s) used

- vidya-karana RAG cache `day-01-intro.json` for the general framing.
- Plofker (2009) for the zero/Bakhshali reference.

Day 2 — The Four Vedas

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Students name the four Vedas, distinguish the four textual layers within each Veda, and identify what counts as “Vedic intellectual contribution” beyond ritual.

Materials

- Wall chart: 4 Vedas × 4 layers grid
- Printed extract: Ṛg Veda 1.1 (Agni hymn) with translation

Vocabulary

- *Samhitā* — “collection”; the hymn layer
- *Brāhmaṇa* — ritual commentaries
- *Āraṇyaka* — “forest texts”; intermediate philosophical
- *Upaniṣad* — “sitting near”; philosophical dialogues

Flow

Warm-up (5 min)

Daily prompt — 2 students share.

Core (20 min)

1. The Four Vedas (write on board):

Veda	Focus
<i>Ṛg Veda</i>	Hymns of praise (~1500–1200 BCE)
<i>Yajur Veda</i>	Ritual formulas
<i>Sāma Veda</i>	Chants set to melody
<i>Atharva Veda</i>	Spells, incantations, early medical

2. Four layers within each Veda:

Samhitā (hymns)
↓
Brāhmaṇa (ritual commentary)
↓
Āraṇyaka (intermediate philosophy)
↓
Upaniṣad (philosophical dialogues – “Vedānta” = “end of the Vedas”)

3. The intellectual contributions (write on board):

- **Sanskrit phonetics & grammar** — the Vedas demanded precise oral transmission, generating sophisticated linguistic analysis (peaking in Pāṇini, ~4th c. BCE).
 - **Early astronomy** — calendar calculation embedded in ritual texts.
 - **Atharva Veda medicine** — the earliest documented Indian medical material; Ayurveda’s roots.
 - **Upanishadic philosophy** — abstract metaphysical inquiry (later picked up by Sāṅkhya, Vedānta, etc.).
4. **Read aloud:** opening of *R̥g Veda 1.1* (the Agni hymn) — 4 lines in Sanskrit + English.
- Note: hymn form, ritual purpose, also a window into 1500 BCE worldview.

Activity (15 min) — Layer mapping

Students draw their own version of the “4 Vedas × 4 layers” grid in their notebook. For each cell, they write one one-line description.

Then they pick ONE cell and write a 1-sentence answer to: “*What might someone studying THIS layer be trying to understand?*”

Wrap-up (5 min)

- Quick exit ticket (oral): “Name one Veda and one layer.”
- Preview Day 3: “*Tomorrow — the Vedangas, six auxiliary disciplines that grew up around the Vedas.*”

Student-facing content

The Four Vedas

Veda	What it contains
R̥g	Hymns of praise (~1500–1200 BCE)
Yajur	Ritual formulas
Sāma	Chants and melody
Atharva	Spells, early medicine

Each Veda has FOUR layers: **Samhitā** (hymns) → **Brāhmaṇa** (ritual commentary) → **Āraṇyaka** (forest texts) → **Upaniṣad** (philosophical dialogues).

The *intellectual* contributions of the Veda include: Sanskrit grammar, early astronomy, early medicine, and Upanishadic philosophy.

Homework

- Find ONE Upaniṣad name (e.g. Īśa, Kena, Kaṭha, Praśna, Muṇḍaka, Māṇḍūkya — there are 108 traditionally, with 13 considered “principal”). Bring tomorrow.

Differentiation

- **Tier up:** Read the Īśa Upaniṣad in translation (it's 18 verses; the shortest principal Upaniṣad).
- **Tier down:** Just remember the four Vedas + the four layers.

Teacher notes

- The dating “~1500–1200 BCE” for the Ṛg Veda is the mainstream scholarly view (Indo-European migration model). Some Indian scholars argue for earlier dates. Acknowledge both; this isn't the module to resolve the debate.
- The four-Veda framing is itself a classification. Some Vedic texts don't fit cleanly. The grid is a useful first map, not the final word.

Citation(s) used

- Ṛg Veda 1.1 — Jamison & Brereton (2014) translation.
- Pollock (2006) on Vedic transmission and linguistic precision.

Day 3 — The Six Vedangas

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Students name the six Vedāṅgas (auxiliary disciplines) and match each to its function in preserving and applying Vedic knowledge.

Materials

- Whiteboard
- Matching-game cards (6 names + 6 functions, shuffled)

Vocabulary

- *Vedāṅga* — “limb of the Veda”
- *vyākaraṇa* — “grammar; analysis”
- *jyotiṣa* — “luminary; astronomy/astrology”

Flow

Warm-up (5 min)

Daily prompt + recall: 2 students name the 4 Vedas + 4 layers from yesterday.

Core (20 min)

1. **The Six Vedāṅgas** (write on board, then build each row):

#	Sanskrit	Function	Modern equivalent
1	<i>Śikṣā</i>	Phonetics, pronunciation	Linguistics / phonology
2	<i>Kalpa</i>	Ritual procedure	Liturgy
3	<i>Vyākaraṇa</i>	Grammar (peaks in Pāṇini ~4th c. BCE)	Formal grammar / parsing
4	<i>Nirukta</i>	Etymology	Lexicography
5	<i>Chandas</i>	Prosody (poetic metre)	Poetics
6	<i>Jyotiṣa</i>	Astronomy / time-keeping	Astronomy + astrology

2. **Why they exist.** The Vedas had to be transmitted PRECISELY, orally, across millennia. Six disciplines grew up around the texts to ensure correct pronunciation, ritual use, grammatical understanding, etymological clarity, metrical recitation, and astronomical timing.

3. **One stand-out: Pāṇini's *Aṣṭādhyāyī* (~4th c. BCE).**

- ~4000 sūtras describing Sanskrit grammar.
- Considered the first known formal grammar of any language.
- Influenced modern formal-language theory (Chomsky's generative grammar acknowledges Pāṇinian inspiration).
- A single Vedāṅga (vyākaraṇa) produced one of the most important works in the history of linguistics.

4. **One more stand-out: Jyotiṣa.**

- The astronomy/astrology pair. Distinguish: jyotiṣa contains BOTH calculation (astronomy) and interpretation (astrology). Modern science separates these; the Indian tradition kept them together.

Activity (15 min) — Matching game

Pairs. 12 cards (6 Sanskrit names + 6 function descriptions) shuffled face-down. Memory-match game. First pair done writes their match on the board.

Then class discusses: “Which Vedāṅga surprised you most? Why?”

Wrap-up (5 min)

- Exit ticket: name 3 of the 6 Vedāṅgas with their functions.
- Preview Day 4: “Tomorrow — the Upavedas. Four APPLIED sciences.”

Student-facing content

The Six Vedāṅgas — “limbs of the Veda”

#	Sanskrit	Function
1	Śikṣā	Pronunciation, phonetics
2	Kalpa	Ritual procedure
3	Vyākaraṇa	Grammar
4	Nirukta	Etymology
5	Chandas	Poetic metre
6	Jyotiṣa	Astronomy & astrology

Pāṇini's Aṣṭādhyāyī (~4th c. BCE) — under Vyākaraṇa — is widely considered the first formal grammar of any language. ~4000 sūtras.

Homework

- Look up ONE Sanskrit word in an online etymology resource. Note where it might appear in English vocabulary (e.g. *mātr* → mother). Bring tomorrow.

Differentiation

- Tier up:** Read about the Chomsky-Pāṇini connection — find 1 specific example of Pāṇinian influence on modern linguistics.
- Tier down:** Memorise just 3 Vedāṅgas: Vyākaraṇa, Jyotiṣa, Kalpa.

Teacher notes

- Students often confuse Jyotiṣa with horoscope-astrology. Distinguish: the discipline contained BOTH planetary calculation AND interpretive use. Modern science separates these.
- The “first formal grammar” claim about Pāṇini is mainstream scholarly view (Cardona, Kiparsky, etc.). Defensible.

Citation(s) used

- Pollock (2006) on Vedic disciplines.
- Cardona, G. (1997). *Pāṇini: His Work and Its Traditions*. Motilal Banarsidass.

Day 4 — The Four Upavedas

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Students name the four Upavedas and one specific contribution of each.

Materials

- Whiteboard
- Visual cards or images (a surgical instrument, a vīṇā, a martial pose, a manuscript)

Vocabulary

- *Upaveda* — “subordinate Veda”; applied sciences
- *cikitsā* — “treatment, therapy”

Flow

Warm-up (5 min)

Daily prompt + recall: name 3 of the 6 Vedāṅgas.

Core (20 min)

1. The Four Upavedas — the APPLIED sciences:

#	Sanskrit	Domain	Primary text
1	<i>Āyurveda</i>	Medicine	Caraka Saṃhitā (~2nd c. BCE), Suśruta Saṃhitā
2	<i>Dhanurveda</i>	Martial arts & warfare	Various manuals; partial extant
3	<i>Gandharvaveda</i>	Music	Bharata’s <i>Nāṭyaśāstra</i> (~2nd c. BCE)
4	<i>Arthaśāstra</i>	Governance & economics	Kauṭilya’s <i>Arthaśāstra</i> (~3rd c. BCE)

(Note: classifications vary. *Sthāpatya-veda* — architecture — sometimes counted as a 5th. Stick to the standard 4 for clarity.)

2. Tour each briefly:

Āyurveda — *Caraka Saṃhitā* lays out a full theory of health, the three doshas (vāta, pitta, kapha), diagnostic methods, and pharmacology. Module 3 covers this in depth.

Dhanurveda — From *dhanus* (bow). Military training, weapon usage, fortification. Many texts lost; what survives is referenced in epics and later manuals.

Gandharvaveda — Music and performing arts. Bharata’s *Nāṭyaśāstra* covers theatre, music, dance — the foundational text for Indian classical performing arts.

Arthaśāstra — Kauṭilya (a.k.a. Cāṇakya) wrote a treatise on statecraft, economics, and governance. Compared sometimes to Machiavelli’s *Prince*, but ~1800 years older and more systematic.

3. A specific Upaveda achievement to remember:

- **Suśruta’s surgery** (~6th c. BCE or earlier): the *Suśruta Saṃhitā* describes 300+ surgical procedures including cataract surgery and rhinoplasty (plastic surgery on the nose). The “Sushruta nose” reconstruction technique was learned by European surgeons via translation in the 18th century.

Activity (15 min) — Match an Upaveda to a problem

Teacher gives 5 modern problems on the board. Students decide which Upaveda's tradition would have addressed each:

1. “I have a chronic skin issue, what would the tradition suggest?” (Āyurveda)
2. “Our city needs a new defence strategy” (Dhanurveda + Arthaśāstra)
3. “How should taxes be calculated for the rainy season?” (Arthaśāstra)
4. “I want to learn a new rāga on the vīṇā” (Gandharvaveda)
5. “Our king wants to build a court of judges” (Arthaśāstra)

Discuss: where would Suśruta have fit? (Āyurveda — surgery is a subdomain of medicine in this scheme.)

Wrap-up (5 min)

- Preview Day 5: “*Tomorrow — mid-module quiz on everything we’ve covered so far.*”

Student-facing content

The Four Upavedas — the APPLIED sciences

#	Sanskrit	Domain
1	Āyurveda	Medicine
2	Dhanurveda	Martial arts
3	Gandharvaveda	Music
4	Arthaśāstra	Governance & economics

Standout achievement: **Suśruta's surgical procedures** (~6th c. BCE), including cataract surgery and nose reconstruction.

Homework

- Look up “Kauṭilya's Arthaśāstra” online. Find ONE specific governance idea it contains (e.g. spy networks, foreign-policy doctrine, tax classifications). Bring tomorrow.

Differentiation

- **Tier up:** Read the first chapter of the *Nāṭyaśāstra* in translation. What does it claim is the *origin* of theatre?
- **Tier down:** Memorise just the four Upavedas with one-line domains.

Teacher notes

- The classification of Upavedas is somewhat traditional and not perfectly clean. Some sources include 5 (adding Sthāpatya-veda — architecture). Stick to 4 for clarity; mention the 5th briefly.

- The Sushruta-nose-reconstruction history is genuine; European surgeons did learn the technique via 18th-century Bengali practitioners. It's a documented case of pre-modern Indian medical knowledge influencing modern medicine.

Citation(s) used

- Wujastyk (2003) on Caraka & Suśruta.
- Boesche, R. (2003). *The First Great Political Realist: Kautilya and His Arthashastra*. Lexington.
- Bharata, *Nāṭyaśāstra* — various editions.

Day 5 — Mid-Module Check

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Students consolidate Days 1–4 via a formative quiz and commit to a project topic.

Materials

- Formative quiz Part B (`quizzes/formative.md`)
- Project brief handouts
- Topic sticky notes

Flow

Warm-up (5 min)

Daily prompt. Brief recap chorus: 4 Vedas, 4 layers, 6 Vedangas (just names).

Formative quiz (15 min)

- `quizzes/formative.md` Part B.
- Closed notebook. 10 questions covering Vedas / layers / Vedangas / Upavedas / one named contribution.

Review (10 min)

- Class walk-through of common errors. Don't grade now; mark tonight.

Project commitment (10 min)

- Hand out project brief.
- Each student picks ONE strand:
 - Mathematics (zero, decimal place-value, infinite series, Bhāskara, Brahmagupta, Mādhava)
 - Astronomy (Aryabhata, Sūrya-siddhānta)
 - Medicine (Caraka, Suśruta, Āyurveda doshas)

- Linguistics (Pāṇini's grammar)
- Music (*Nāṭyaśāstra*, rāga system)
- Architecture / Sthāpatya
- Governance (Kauṭilya)
- Other (with teacher approval)
- Write topic on a sticky note + post on the wall under their chosen domain.

Wrap-up (5 min)

- Note: "Some domains will have 5+ students, some will have 1. Each will be different. That's fine."
- Preview Day 6: "*Tomorrow — deep on Indian mathematics. Zero. Place value. Infinite series.*"

Homework

- Submit project plan paragraph (working title + thesis sentence + 2 sources) by tomorrow.

Differentiation

- **Tier up:** Pick a less common topic (Pāṇini, Mādhava). The research is harder; the project will stand out.
- **Tier down:** Pick a well-documented topic (zero, Aryabhata, Suśruta). Easier research.

Teacher notes

- Mark the formative quiz tonight. Note any student who scored below 50% — schedule a 10-min check-in by Day 7.
- The project topic choice often surfaces what students are curious about. Adjust the depth of Days 6 (math) and 7 (astronomy) based on the distribution.

Citation(s) used

- All sources covered to date.

Day 6 — Indian Mathematics

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Students name 3 specific Indian mathematical contributions with dates and articulate why each matters.

Materials

- Whiteboard
- Printed extract: 1 verse from Bhāskara II's *Līlāvati* (12th c. CE) in translation

Vocabulary

- *gaṇita* — “calculation, mathematics”
- *śūnya* — “zero, void”
- *aṅka* — “digit”

Flow

Warm-up (5 min)

Daily prompt + recap: 2 students share one math fact they remember about India from prior knowledge.

Core (25 min)

1. Three specific contributions to remember:

(a) Zero (śūnya) as a number — Bakhshali manuscript (~7th c. CE — older fragments possibly earlier). Distinct from “zero as placeholder” (which existed earlier in Babylonian and Maya systems). The Indian innovation: zero participates in arithmetic operations (you can add, subtract, multiply, divide... with zero).

(b) Decimal place-value system — by 6th c. CE the Indian numeral system (with positional values for ones, tens, hundreds, etc.) was in widespread use. Adopted by Arab mathematicians (Al-Khwārizmī, ~825 CE), reached Europe by ~12th c. via Fibonacci (1202 CE).

(c) Infinite series (centuries before Newton) — Mādhava of Saṅgamagrāma (~14th c. CE, Kerala school) computed infinite series for π , sine, cosine. Specifically:

$$\pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right)$$

(This is now called the “Leibniz series” but Mādhava had it 300 years earlier.)

2. A named text — Bhāskara II’s *Līlāvātī* (1150 CE). A textbook on arithmetic and algebra written for his daughter (the name “*Līlāvātī*” is reportedly hers). Includes problems on geometry, sequences, equations, mensuration. Read one short problem aloud (in translation).

3. What modern math doesn’t credit Indian sources for (but should):

- The number zero as a number (not just placeholder).
- The “Leibniz” series for π (Mādhava had it).
- Several trigonometric identities.

4. What Indian math DIDN’T do:

- Did not develop full coordinate geometry (that’s Descartes, 17th c. European).
- Did not develop calculus formally (though Mādhava came very close with the series).

- Did not develop modern algebraic notation.
- Both lists matter for honesty.

Activity (10 min) — Compute the Mādhava series

Class together. On the board: - Start at 1. - Subtract 1/3. Add 1/5. Subtract 1/7. Add 1/9. ... - After 4 terms: $1 - 0.333 + 0.2 - 0.143 = 0.724$ - Multiply by 4: 2.896 (close to $\pi = 3.14$) - After 10 terms it gets closer.

The series converges SLOWLY — that’s one reason it wasn’t useful as a calculation method. But proving the *form* was a huge mathematical achievement.

Wrap-up (5 min)

- Preview Day 7: “Tomorrow — Indian astronomy. Aryabhata.”

Student-facing content

Three Indian Mathematical Contributions

1. **Zero as a number** — Bakhshali manuscript, ~7th c. CE.
2. **Decimal place-value system** — by 6th c. CE; adopted by Arab mathematicians (~825 CE); reached Europe (~12th c.).
3. **Infinite series for π** — Mādhava of Saṅgamagrāma, ~14th c. CE.

$$\pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right)$$

(Now sometimes called the “Leibniz series” — Mādhava had it ~300 years before Leibniz.)

A named text: **Bhāskara II’s *Līlāvātī*** (1150 CE).

Homework

- Compute 10 terms of the Mādhava series for π . How close do you get?

Differentiation

- **Tier up:** Read about the Madhava-Leibniz attribution debate. Why is the European name still standard?
- **Tier down:** Just remember the three contributions + their approximate dates.

Teacher notes

- The Leibniz-Mādhava attribution is a recurring honest-credit issue. Mainstream math history is increasingly correcting this; some textbooks now use “Madhava-Leibniz series.”
- Bhāskara II’s *Līlāvātī* problems are charming — many are presented as little stories. If time permits, read one or two more.

Citation(s) used

- Plofker (2009), Chapters 4–7.
- Bhāskara II, *Līlāvati* (1150 CE) — translation by Colebrooke (1817) or Patwardhan, Naimpally, Singh (2001).
- Hayashi (1995) on the Bakhshali manuscript.

Day 7 — Indian Astronomy

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Students name 3 specific Indian astronomical contributions with their authors/dates and articulate what each got right and what it got wrong.

Materials

- Whiteboard
- Printed extract: 1 verse from *Aryabhaṭīya* in translation
- Globe (if available) for the rotation demo

Vocabulary

- *jyotiṣa* — astronomy/astrology (Vedanga #6)
- *siddhānta* — “established treatise; systematic astronomical text”
- *nakṣatra* — “lunar mansion; star group”

Flow

Warm-up (5 min)

Daily prompt + recall: 2 students name one Vedanga.

Core (25 min)

1. Three specific contributions:

(a) Aryabhata (c. 476–550 CE) — wrote *Aryabhaṭīya* at age 23.

- Proposed the **earth rotates on its axis** (this is correct). Most contemporaries believed the sky rotated.
- Approximated **π to four decimal places** (3.1416).
- Computed planetary periods and eclipse predictions.
- Did NOT propose heliocentrism (the sun-centred model is Copernicus, 1500s).

(b) Sūrya-siddhānta (early CE) — comprehensive astronomical text.

- Detailed planetary calculations.
- Earth’s circumference computed with ~1% accuracy.

- Eclipse predictions accurate enough to publish almanacs.

(c) Varāhamihira (~6th c. CE) — *Pañca-siddhāntikā*. Compares 5 different astronomical systems including a Greek-derived one. Demonstrates that 6th-c. Indian astronomers were actively engaging with Greek (Yavana) astronomical work.

2. What Indian astronomy DID well:

- Precise calendar (lunar + solar reconciliation).
- Eclipse prediction (eclipses were correctly understood as caused by occlusion, not by the demon Rāhu — though the mythological framing co-existed).
- Trigonometric tables (sines, not chords as in Greek tradition).

3. What it DIDN'T do:

- Did not develop heliocentrism (despite Aryabhata's hint, the geocentric model dominated until modern times).
- Did not develop telescopic observation (no telescope until the 17th c. anywhere).

4. The interesting in-between: the Indian astronomical tradition kept astronomy and astrology TOGETHER under jyotiṣa. Calculation skills used to compute eclipses were the same skills used to compute horoscopes. Modern science separates these; the Indian tradition did not.

Activity (10 min) — Aryabhata's earth-rotation argument

Demonstrate with a globe (or a ball with a marked point): - Set the ball spinning. From the spinner's point of view, the room "rotates" around them. - Aryabhata reasoned: maybe the stars don't rotate around us; maybe WE rotate. - This was the correct answer. He was a thousand years ahead of European thinkers on this specific point.

Wrap-up (5 min)

- Preview Day 8: "Project work begins tomorrow."

Student-facing content

Three Indian Astronomical Contributions

1. **Aryabhata (~500 CE)** — proposed earth rotates on its axis; computed π to 4 decimals; predicted eclipses.
2. **Sūrya-siddhānta (early CE)** — planetary calculations + earth's circumference to ~1% accuracy.
3. **Varāhamihira (~6th c. CE)** — *Pañca-siddhāntikā* compared 5 astronomical systems including Greek-derived ones.

Aryabhata's earth-rotation claim was ~1000 years ahead of European thinkers on that specific point.

Note: Indian astronomy did NOT develop heliocentrism (Copernicus, 1500s) or telescopes (17th c.).

Homework

- Finalise project plan. Bring all materials tomorrow.

Differentiation

- **Tier up:** Read about the Yavana (Greek-Indian) interactions in astronomy. Was the influence one-way?
- **Tier down:** Memorise three names with one fact each.

Teacher notes

- The “Aryabhata predicted earth rotation” claim is mainstream and well-supported. Don’t oversell — he didn’t propose heliocentrism.
- The Yavana-influence question is rich: Indian astronomy genuinely engaged Greek astronomy via the Pañca-siddhāntikā. Both traditions exchanged ideas. Honest history.

Citation(s) used

- Aryabhaṭīya (499 CE) — Clark (1930) translation.
- Plofker (2009), Chapter 4.
- Pingree, D. (1996). “Astronomy in India” in *Astronomy Before the Telescope*.

Day 8 — Project Session 1: Research and Outline

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Each student commits to a specific topic, drafts a thesis sentence, and outlines a 5-minute presentation.

Materials

- Library books / curated reading list
- Project brief + rubric

Flow

Warm-up (5 min)

Brief: today’s project work. Show the rubric on the wall. Cover the key criteria.

Core sprints (35 min)

Sprint 1 (10 min) — Thesis. Each student writes: - Working title - One-sentence thesis (specific claim about their topic) - 2 sources

Teacher rotates: 60 sec per student, asking “*What’s the specific claim? Not ‘X is interesting’ — what specifically are you arguing?*”

Sprint 2 (10 min) — Outline. 5-slide / 5-section outline: 1. The topic + named practitioner 2. What it claimed / did 3. One specific source (primary verse, theorem, or text) 4. Modern view / parallel 5. Honest limitation (what the tradition didn't do)

Sprint 3 (15 min) — Begin slides / notes. Each student starts drafting their actual presentation content.

Wrap-up (5 min)

- Plan submitted at door.
- Preview Day 9: *“Tomorrow — rehearse and refine.”*

Homework

- Continue drafting at home. Tomorrow is for *refining*, not starting.

Differentiation

- **Tier up:** Add a counter-argument: “Some scholars dispute X” with citation.
- **Tier down:** Use the 5-section outline as your full structure.

Teacher notes

- The “specific claim” filter is the most useful intervention. Many students start with “Indian math is interesting” — push to “Mādhava’s series converges to π but converges slowly; here’s why that mattered.”

Day 9 — Project Session 2: Refine and Rehearse

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Each student rehearses their 5-min presentation end-to-end with timed feedback from a buddy.

Materials

- All in-progress materials
- Stopwatch

Flow

Warm-up (3 min)

Brief: today is refine + rehearse.

Sprints (35 min)

Sprint 1 (15 min) — Refine. Teacher rotates with focused question: *“Show me your strongest 60 seconds.”*

Common interventions: - Missing the “honest limitation” — must include. - Source cited but not actually engaged — push for a quote or paraphrase. - Specific claim vague — sharpen.

Sprint 2 (15 min) — Buddy rehearsal. Pairs rehearse to each other. 5 min talk + 2 min feedback. Swap.

Sprint 3 (5 min) — Adjust. Each student decides one change.

Wrap-up (7 min)

- 3 students share what they’re changing.
- Final touches at home tonight.

Homework

- Finalise. Bring everything tomorrow.

Differentiation

- **Tier up:** Anticipate the toughest audience question and prepare a 30-sec answer.
- **Tier down:** Read from notes during presentation if needed.

Teacher notes

- The buddy rehearsal is the most useful feedback. Encourage *kind and specific* feedback.

Day 10 — Final Presentations + Summative

Module: What is IKS? (Middle) · **Time:** 60 min (extend if possible)

Learning objective

Students demonstrate mastery via 5-min presentations + short summative quiz.

Materials

- Summative quiz (quizzes/summative.md)
- Rubric copy per student
- Stopwatch

Flow

Quiz (15 min)

20 marks: MC + short answer + matching + one open response.

Presentations (40 min)

- 5 min talk + 1 min Q&A per student.
- ~6–7 students fit. Book a second period for the rest.
- During each:

- Teacher marks rubric in real time.
- 1 audience question per presentation.
- “I learned / I wonder” slips for the presenter.

Wrap-up (5 min)

- Closing question: “Pick the ONE Indian contribution you’ll keep talking about beyond this class.”
- Reflection slip.

Homework

- None. Module complete.

Differentiation

- **Tier up:** Early-finishers help peer-edit a struggling student’s slide.
- **Tier down:** Open-notebook quiz for documented learning needs.

Teacher notes

- Mark quizzes within 48 hours.
- Capture photos of best slides for the BACKLOG of “exemplars” for next year.

Writing Prompts

Writing Prompts — Module 1 (Middle)

1. **Reflection (Day 1).** Before this module, what did you think “Indian Knowledge” meant? In 4–5 sentences, how has that changed?
2. **Reflection (Day 4).** Pick one Upaveda. Which problem in your own life could it speak to?
3. **Explanatory (Day 6, ~150 words).** Explain to a friend who has never heard of it: what is the Bakhshali manuscript, and why does it matter?
4. **Explanatory (Day 7, ~150 words).** What did Aryabhata get right? What did he NOT propose (despite the popular claim)?
5. **Argumentative (Day 5 or homework).** Some scholars argue “IKS” is too broad a category to be useful. Defend or refute in 1 page with one example.
6. **Creative (Day 4).** Write a 1-paragraph imaginary letter from Kauṭilya (the Arthaśāstra author) to a modern Indian Prime Minister.
7. **Cross-disciplinary.** Pick a contemporary issue (climate, public health, AI ethics). Which Upaveda’s tradition has something to say? Defend.

Homework

Homework — Module 1 (Middle)

Day	Task	Time
1 → 2	Read the project brief. Think about which strand interests you.	15 min
2 → 3	Find one Upaniṣad name (look up the list of 13 principal ones).	10 min
3 → 4	Look up one Sanskrit etymology online. Bring an example.	10 min
4 → 5	Look up “Kauṭilya’s Arthaśāstra” — find one specific governance idea.	15 min
5 → 6	Submit project plan (working title + thesis sentence + 2 sources).	30 min
6 → 7	Compute 10 terms of the Mādhava π series.	15 min
7 → 8	Finalise topic; gather research materials.	30 min
8 → 9	Continue drafting at home (do NOT start tomorrow’s class without progress).	30–45 min
9 → 10	Final touches. Practise once aloud.	15 min